

Introduction to Programming

Lecture 2 — Reproducing tasks with functions

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Université de Saint-Étienne | Year 2026/2027

-1 **Recap' time**

Recap: Variables

Recap: Variables

- Have a type
- Have a declaration and initialization
- Store a value
- Have a scope
- Can be constant with `const`

Recap: Booleans

Recap: Booleans

- Have type `bool`
- Can only have one two values: `true` or `false`
- Can be combined with operations like `&&` (logical and) and `||` (logical or)
- The result of comparison expressions is a boolean

Recap: Conditional statements

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- Using `if`, `else if`, `else` or `switch-case/default`
- Allow branching according to the value of a boolean expression or variable

Recap: Loops

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- Conditional loops with `while`
- Finite loops with `for`
- Allow repeating actions

Rewind: Conditional Loops

Repeat an action as long as the condition is `true`.

```
while(a_condition) {  
    // code to repeat  
}
```

Rewind: Conditional Loops

Repeat an action as long as the condition is **true**.

```
while(a_condition) {  
    // code to repeat  
}
```

```
int i = 0;  
while(i < 10) {  
    // increase i by one as long  
    as i < 10  
    ++i;  
}
```

Rewind: Sequential Loops

Repeat an action an known amount of time.

```
for(auto counter = 0; counter < limit; counter++) {  
    // counter is incremented by one at the end of the block  
}  
// counter only exists inside the for-loop block
```

Rewind: Sequential Loops

Repeat an action an known amount of time.

```
for(auto counter = 0; counter < limit; counter++) {  
    // counter is incremented by one at the end of the block  
}  
// counter only exists inside the for-loop block  
  
const int upper_bound = 10;  
for(auto counter = 0; counter < upper_bound; counter++) {  
    std::cout << counter << " ";  
}
```

00

Assignment review

Solution: Exercise 1

Let's look at the solution: <https://godbolt.org/z/cEeT3x9Kr>

Solution: Exercise 2

Let's look at the solution: <https://godbolt.org/z/T9rq1PvYM>

01

**Recycle pieces of code
with Functions**

Fresh example

A brand new example

About functions

Functions allow reusing portions of code. They have:

About functions

```
void print();
```

Functions allow reusing portions of code. They have:

- A declaration

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Functions allow reusing portions of code. They have:

- A declaration
- A definition

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void print();
```

```
void print() {  
    std::cout << "Hello, World!" <<  
        std::endl;  
}
```

About functions

Functions allow reusing portions of code. They have:

- A declaration
- A definition

‘Calling a function’ means using it.

```
void print();
```

```
void print() {  
    std::cout << "Hello, World!" <<  
        std::endl;  
}
```

```
print();
```

02 Parameterizing functions

Arguments

Arguments

Functions can take **arguments**. It's called *passing argument(s)* to a function.

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```
void print(std::string message) {  
    std::cout << message << std::endl;  
}
```

Arguments

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```
void print(std::string message) {  
    std::cout << message << std::endl;  
}  
  
void two_lines(std::string line1, std::string line2) {  
    std::cout << line1 << std::endl;  
    std::cout << line2 << std::endl;  
}
```

Arguments

Functions can take **arguments**. It's called *passing argument(s)* to a function.

```
void print(std::string message) {  
    std::cout << message << std::endl;  
}  
  
void two_lines(std::string line1, std::string line2) {  
    std::cout << line1 << std::endl;  
    std::cout << line2 << std::endl;  
}
```

```
print("Hello, World!");  
two_lines("Hello,", "World!");
```

Arguments

Functions can take **arguments**. It's called *passing argument(s)* to a function.

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void print(std::string message) {  
    std::cout << message << std::endl;  
}  
  
void two_lines(std::string line1, std::string line2) {  
    std::cout << line1 << std::endl;  
    std::cout << line2 << std::endl;  
}
```

```
print("Hello, World!");  
two_lines("Hello,", "World!");
```

```
std::string msg1 = "Hello"  
std::string msg2 = "world";  
two_lines(msg1, msg2);
```

Practise time!

Value vs Reference

When passing an argument to a function, you either let that function:

Value vs Reference

When passing an argument to a function, you either let that function:

- copy the argument's value: any modification to the argument done in the function won't be effective outside of it;

```
void plus_one(int value) {  
    value = value + 1;  
    std::cout << value << std::endl;  
}  
  
int myvar = 4;  
plus_one(myvar);  
std::cout << myvar << std::endl; // prints 4
```

Value vs Reference

When passing an argument to a function, you either let that function:

- take the argument's reference: modification of the variable in the function's scope will be effective outside of it.

```
void plus_one(int& value) {
    value = value + 1;
    std::cout << value << std::endl;
}

int myvar = 4;
plus_one(myvar);
std::cout << myvar << std::endl; // prints 5
```

Passing by value, in more depth

When passing by value, a local variable is created in the function, and initialized with the given argument's value.

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void plus_one(int local_var) {  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

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When passing by value, a local variable is created in the function, and initialized with the given argument's value.

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void plus_one(int local_var) {  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

```
int myvar = 4;  
plus_one(myvar);
```

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When passing by value, a local variable is created in the function, and initialized with the given argument's value.

```
void plus_one(int local_var) {
    local_var = local_var + 1;
    std::cout << local_var << std::endl;
}
```

```
void plus_one(int local_var) { // local_var
    initialized with value of myvar
```

```
int myvar = 4;
plus_one(myvar);
```

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When passing by value, a local variable is created in the function, and initialized with the given argument's value.

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void plus_one(int local_var) {  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

```
int myvar = 4;  
plus_one(myvar);
```

```
void plus_one(int local_var) { // local_var  
    initialized with value of myvar  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

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void plus_one(int local_var) {  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

```
int myvar = 4;  
plus_one(myvar);
```

```
void plus_one(int local_var) { // local_var  
    initialized with value of myvar  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
} // local_var is deleted  
// but myvar subsists
```

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}
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    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

```
int myvar = 4;  
plus_one(myvar);
```

Passing by reference, in more depth

When passing by reference, the function's variable is like an alias of the given argument.

```
void plus_one(int& local_var) {  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

```
void plus_one(int& local_var) { // local_var  
    and myvar denote the same thing
```

```
int myvar = 4;  
plus_one(myvar);
```

Passing by reference, in more depth

When passing by reference, the function's variable is like an alias of the given argument.

```
void plus_one(int& local_var) {  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

```
int myvar = 4;  
plus_one(myvar);
```

```
void plus_one(int& local_var) { // local_var  
    and myvar denote the same thing  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

Passing by reference, in more depth

When passing by reference, the function's variable is like an alias of the given argument.

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void plus_one(int& local_var) {  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
}
```

```
int myvar = 4;  
plus_one(myvar);
```

```
void plus_one(int& local_var) { // local_var  
    and myvar denote the same thing  
    local_var = local_var + 1;  
    std::cout << local_var << std::endl;  
} // the symbol local_var is deleted  
// myvar subsists, and its value changed 14/37
```

Practise time!

Default arguments 1/3

Some arguments can get a *default* value. If the caller doesn't provide a value for the default argument, it will take its default value. Otherwise, the argument takes the user-provided value.

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void clamp(int pixel, int low = 0, int high = 25);
```

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```
clamp(r); // low = 0, high = 255
```

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void clamp(int pixel, int low = 0, int high = 25);
```

```
clamp(r); // low = 0, high = 255
```

```
clamp(r, 10); // low = 10, high = 255
```

Default arguments 1/3

Some arguments can get a *default* value. If the caller doesn't provide a value for the default argument, it will take its default value. Otherwise, the argument takes the user-provided value.

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void clamp(int pixel, int low = 0, int high = 25);
```

```
clamp(r); // low = 0, high = 255
```

```
clamp(r, 10); // low = 10, high = 255
```

```
clamp(r, 0, 128); // low = 0, high = 128
```

Default arguments 2/3

Default arguments are positionned last in the list of arguments. Consider :

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```
void foo(int arg1, int default1 = 9, int arg2);
```

Default arguments 2/3

Default arguments are positionned last in the list of arguments. Consider :

```
void foo(int arg1, int default1 = 9, int arg2);  
foo(1,2); // who takes the value 2? default1 or arg2 ?  
// ambiguous, so we forbid it
```

Default arguments 3/3

Default arguments are only specified in the function's declaration.

If you split the declaration from the definition, you don't have to repeat the arguments default values:

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If you split the declaration from the definition, you don't have to repeat the arguments default values:

```
void score_song(std::string song_name, bool verbose =  
    False);
```

```
void score_song(std::string song_name, bool verbose) {  
    // verbose is still a default argument  
    // function body here  
}
```

Little practise time!

03

**Getting results from
functions**

With references

With references

So far, you can get information back from a function through arguments passed by reference:

```
void plus(int a, int b, int& res) {  
    res = a + b;  
}
```

```
int res;  
plus(1,2,res);  
std::cout << res; // prints 3
```

Function's return value

Functions can *return* a value.
Let's look at that example

Function's signature

When declaring a function, you are declaring its *signature*.

```
return_type function_name (arg1_type argument1, ... );
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```
void print (const std::string message);
```

Function's signature

When declaring a function, you are declaring its *signature*.

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return_type function_name (arg1_type argument1, ... );
```

```
int multiply (int a, int b);
```

```
void print (const std::string message);
```

```
int random ();
```

Functions's exit point(s)

To return a value from a function, you use the `return` directive.

You can return from a function at different points in your program.

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```
std::string genre_label(const Genre genre){
    switch(genre){
        case Genre::Jazz:
            return "Jazz";
        case Genre::Rock:
            return "Rock";
        case Genre::Classical:
            return "Classical";
        default:
            return "Unknown";
    }
}
```

Functions early exit

One can return in the middle of a function, with a conditional statement, and/or during a loop.

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One can return in the middle of a function, with a conditional statement, and/or during a loop.

```
int complicated_get_4(int low = 0, int high = 10){  
    for(auto number = 0; number < 10; ++number){  
        if(number == 4)  
            return number;  
    }  
    return 4;  
}
```

Functions early exit

One can return in the middle of a function, with a conditional statement, and/or during a loop.

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int complicated_get_4(int low = 0, int high = 10){  
    for(auto number = 0; number < 10; ++number){  
        if(number == 4)  
            return number;  
    }  
    return 4;  
}  
complicated_get_4();
```

Functions early exit

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int complicated_get_4(int low = 0, int high = 10){  
    for(auto number = 0; number < 10; ++number){  
        if(number == 4)  
            return number;  
    }  
    return 4;  
}  
  
complicated_get_4();  
complicated_get_4(7,12);
```

Functions early exit

One can return in the middle of a function, with a conditional statement, and/or during a loop.

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int complicated_get_4(int low = 0, int high = 10){
    for(auto number = 0; number < 10; ++number){
        if(number == 4)
            return number;
    }
    return 4;
}

complicated_get_4();
complicated_get_4(7,12);
```

Don't forget **return** in your non-void functions, or your program might crash! 22/37

Practise time!

04

**One function, Multiple
definitions**

Adding numbers

```
int add(int a, int b) { return a+b; }  
add(1, 2);
```

Adding numbers

```
int add(int a, int b) { return a+b; }
```

```
add(1, 2);
```

```
// add(1, 2, 3); ?
```

Solution here

Overloading functions 1/2

A function can have multiple definitions: they're called **overloads**.

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You can *overload* a function by:

- Using a different amount of arguments

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You can *overload* a function by:

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- Changing the arguments types

Overloading functions 1/2

A function can have multiple definitions: they're called **overloads**.

You can *overload* a function by:

- Using a different amount of arguments
- Changing the arguments types
- Using a different amount of arguments with differing types

Overloading functions 2/2

Subtleties:

You can overload a function by:

- By **changing the constness** of arguments that are passed by **reference**

Overloading functions 2/2

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You **cannot** overload a function by:

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Overloading functions 2/2

Subtleties:

You can overload a function by:

- By **changing the constness** of arguments that are passed by **reference**

You **cannot** overload a function by:

- Using arguments passed by **value** vs using arguments passed by **reference**

The idea is to wonder if you'd be able to call a specific overload without ambiguity — that is, you can find a specific situation where only one overload matches.

Overload through constness

You can overload with const references because there will be case where only one overload will be legal.

```
int add(int& a, int& b) {  
    return a + b;  
}  
  
int add(const int& a, const int& b) {  
    return a + b;  
}  
  
const int a = 1, b = 2;  
add(a,b); // ok: a and b are const, calls second overload
```

Overload through constness

But what about cases where both can be used ?

```
int add(int& a, int& b) {  
    return a + b;  
}  
  
int add(const int& a, const int& b) {  
    return a + b;  
}  
  
int a = 1, b = 2;  
add(a,b); // ?
```

Ambiguity of references

The standard will prefer the candidate that is closest to the caller's capability.

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```
int add(int& a, int& b) {  
    return a + b;  
}  
  
int add(const int& a, const int& b) {  
    return a + b;  
}  
  
int a = 1, b = 2; // a and b are mutable  
add(a,b);
```

Ambiguity of references

The standard will prefer the candidate that is closest to the caller's capability.

```
int add(int& a, int& b) {  
    return a + b;  
}
```

```
int add(const int& a, const int& b) {  
    return a + b;  
}
```

```
int a = 1, b = 2;  
add(a,b); // we prefer the first overload because it takes mutable  
           references
```

Practise time!

05

Everything about scope

Local variables

A *local* variable is a variable that is accessible only in the perimeter of a block or function.

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```
void a_function() {  
    int local_var; // local to a_function  
}  
  
int main(){  
    {  
        int local_var; // local in current block  
    }  
    int local_var; // local in main  
}
```

Arguments passed by-value

Arguments passed by value to a function are local variables to that function.

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```
void a_function(int a) {
```

Arguments passed by-value

Arguments passed by value to a function are local variables to that function.

```
void a_function(int a) {  
    // a is declared here and initialized to whatever value was  
    // given to a_function  
}  
  
int main(){  
    int a = 9; // a is local to main  
    a_function(a); // main's a value is copied into a_function's a.  
    // They are two separate spaces in memory.  
}
```

Global variables

A *global* variable is accessible everywhere in the code, from the moment it's been declared. It is declared outside of any function or block.

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```
void foo() {
    // glob is not accessible here
}

const int glob = 10;

void bar () {
    std::cout << glob << std::endl; // ok, glob exists
}

int main() {
    std::cout << glob << std::endl; // ok, glob exists
}
```

Global variables

A *global* variable is accessible everywhere in the code, from the moment it's been declared. It is declared outside of any function or block.

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void foo() {
    // glob is not accessible here
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const int glob = 10;

void bar () {
    std::cout << glob << std::endl; // ok, glob exists
}

int main() {
    std::cout << glob << std::endl; // ok, glob exists
}
```

Global variables are usually used for **constants**.

Variable shadowing

A local variable will *shadow* a global variable that has the same name.

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A local variable will *shadow* a global variable that has the same name.

```
const int glob = 10;
void foo () {
    std::string glob = "Hello"; // local glob is now Hello
    std::cout << glob << std::endl; // prints Hello
}

int main() {
    foo(); // prints hello
    // glob is still 10
    int glob = 9;
    // local glob is 9
    // global glob is still 10
}
```

Accessing shadowed global variable

A global variable that is shadowed can be accessed by appending the prefix `::`.

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A global variable that is shadowed can be accessed by appending the prefix `::`.

```
const int glob = 10;
void foo () {
    std::string glob = "Hello"; // local glob
    std::cout << ::glob << std::endl; // prints 10
}

int main() {
    foo(); // prints 10
    int glob = 9;
    ::glob = 12;
    std::cout << ::glob << std::endl; // 12
    std::cout << glob << std::endl; // 9
}
```

Practise time!

06

The main function

So, `main` is a function, right?

We've covered that `main` is the entry point of your program.

So, `main` is a function, right?

We've covered that `main` is the entry point of your program.
When your program starts, it automatically calls `main`.

Arguments to main

Surprise:

Arguments to main

Surprise: `main` can take arguments.

Arguments to main

Surprise: main can take arguments.

```
int main(int argc, char * argv[]) {  
    return 0;  
}
```

Arguments to main

Surprise: main can take arguments.

```
int main(int argc, char * argv[]) {  
    return 0;  
}
```

Let's see what this means

main's signatures

main can only have one of the two following signatures:

```
int main();
```

```
int main(int argc, char * argv[]);
```

main's signatures

main can only have one of the two following signatures:

```
int main();
```

```
int main(int argc, char * argv[]);
```

There can be only one `main` per program.

Enforcing program usage

Often, when a program expects a specific amount of argument, it will print the way the program is expected to be used when the call didn't provide the right amount of arguments.

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Often, when a program expects a specific amount of argument, it will print the way the program is expected to be used when the call didn't provide the right amount of arguments. The corresponding code is :

```
int main(int argc, char * argv[]) {
    if(argc != 3 ){ // program expects 2 arguments from
        user
        std::cout << "usage: ./program <arg1> <arg2>" <<
            std::endl;
    }
}
```

Optional parameters

Mandatory parameters are generally signaled in between herringbones <>.

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```
int main(int argc, char * argv[]) {  
    if(argc < 3 || argc > 4 ){ // program expects at least 2 and at  
        most 3 arguments from user  
        std::cout << "usage: ./program <arg1> <arg2> [optional1]" <<  
            std::endl;  
    } else if(argc == 4) {  
        // do something special when three arguments are provided  
    }  
}
```

Practise Time!

See you next time!